Assignment 2: Detecting Emotions with ML Algorithms

Cross-Dataset Generalization of Facial Expressions

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5ARE0 – Data Analysis & Learning Methods
Assignment 2 – Group 5
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Example citations for later on (remove this later): (Goodfellow et al., 2013)
(Goeleven et al., 2008)
(Goeleven et al., 2008; Goodfellow et al., 2013)

1 Introduction and research questions

[Replace: Summarize the objective: facial-expression recognition on FER2013 and KDEF, evaluation across domains, explainable fuzzy modeling and practical real-time use.]

The primary question is how datasets and feature/model choices affect within-dataset accuracy and cross-dataset generalization. Secondary questions concern which emotions and representations are most transferable, whether interpretable facial geometry is useful, and how reliable a trained model remains under webcam conditions. *[Replace: Add source-supported motivation and APA citations to FER2013/KDEF after Zotero export.]*

2 Datasets, preprocessing and experimental protocol

FER2013. Images are grayscale, 48×48, with seven labels: angry, disgust, fear, happy, sad, surprise and neutral. Use the provided Training split to fit models, PublicTest for model selection/tuning, and untouched PrivateTest *only* for final evaluation.

KDEF. Use only straight/frontal (S) faces, across seven comparable expressions (KDEF "Afraid" maps to FER "fear"). Divide by identity: 80% of subjects for training, 20% for testing, with no subject overlap.

Common representation. Detect and crop KDEF faces with a margin preserving forehead and jaw; convert to grayscale, resize to 48×48 and apply the same normalization as FER. Apply identical feature extraction at training and inference. Fit any scaling, PCA or learned transformation only on training data.

HOG features. Histogram of Oriented Gradients (HOG) looks at the edges in an image and identifies where the brightness changes and in which direction. To extract the features, the 48×48 grayscale image is cut into a 6×6 grid of cells of 8×8 pixels. In each cell, the change in brightness is measured in 9 different directions $(0°, 20°, \ldots, 160°)$. We then slide a 2×2 block across the grid, which results in 5×5 overlapping block positions:

$$\text{HOG features} = \underbrace{(5 \times 5)}_{\text{block positions}} \times \underbrace{(2 \times 2)}_{\text{cells per block}} \times \underbrace{9}_{\text{directions}} = 900$$

Fuzzy classifier protocol. For Task 4, geometric features were extracted directly from original-resolution frontal KDEF images, rather than the FER-compatible 48×48 images. Both acquisition sessions for each identity remained in the same partition. Model selection used training subjects only, preserving the held-out KDEF subjects for final evaluation.

[Replace: Document dataset counts/class distributions; detected-face failures, deduplication/quality checks, exact train/validation/test protocol, seeds, tools, versions and potential limitations. Provide one dataset example if it supports a claim.]

3 Task 1: FER2013 baseline (20%)

Models and features. Compare Logistic Regression, SVM and Random Forest with HOG, facial landmarks and PCA features. Retain the supplied Random Forest + landmarks starter as a baseline. *[Replace: Describe*

representation dimensions, scaling, PCA explained variance, model configurations and how tuning on PublicTest was controlled.]

Table 1: FER2013 PublicTest (validation) comparison. Fill all nine feature–model combinations; report selected model(s) separately on PrivateTest.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1
HOG	Logistic Regression	–	–	–
HOG	SVM	–	–	–
HOG	Random Forest	–	–	–
Landmarks	Logistic Regression	–	–	–
Landmarks	SVM	–	–	–
Landmarks	Random Forest	–	–	–
PCA	Logistic Regression	–	–	–
PCA	SVM	–	–	–
PCA	Random Forest	–	–	–

Selection and failure analysis. *[Replace: Identify the strongest classifier/features on validation, give final PrivateTest accuracy/balanced accuracy/macro-F1, reference appendix matrices; compare per-emotion recall, class imbalance, hard-to-classify/mislabeled/lighting-varied samples.]*

4 Task 2: KDEF dataset study

In this task we take the best FER2013 representation from Task 1, HOG, retrain the three classifiers on KDEF, and compare the results with FER2013.

The Karolinska Directed Emotional Faces - KDEF was a dataset that was generated in 1998. It has a total of 4900 pictures, of 70 individuals displaying 7 different emotions. The subjects for this dataset was chosen with a specific selection criteria some of which include having no beard, mustache, earrings etc. All subjects were given special gray shirts and all the photos were taken at a distance of 3m from the camera. Each subject was taken photos in 2 sessions, 5 poses for each of the 7 emotions. The pose that is of most interest to us is *straight*

Unlike the FER2013 dataset(which are 48×48 dimensions), the KDEF dataset has images of dimensions 562×762 and RGB images. To compare the 2 datasets, the KDEF images need to be converted to the FER2013 format: a Haar cascade detects the face, the box is enlarged by 20% so the whole face and some context are kept, and the crop is converted to grayscale, resized to 48×48 and scaled to $[0, 1]$

The data is split 80/20 by subject i.e out of the 70 subjects, 56 of them are used in the training set, and 14 are in the test set. This is to ensure there is no data leakage into the test set. KDEF is perfectly balanced, with 112 training and 28 test images per emotion (Figure 4.0.1).

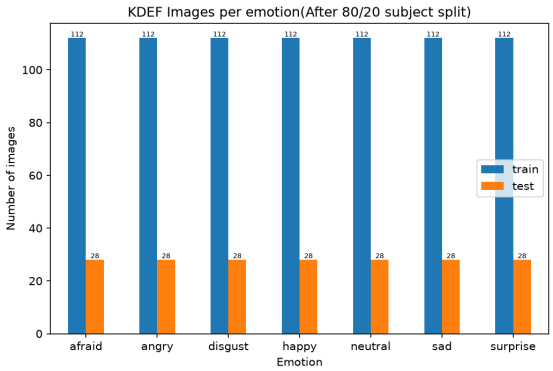


Figure 4.0.1: Number of KDEF images per emotion in the training (56 subjects) and test (14 subjects) split.

In the previous task, we found out that the best feature representation is Histogram of Oriented Gradients (HOG), so every KDEF image is described with the same 900 HOG features (Section 2).

4.1 Model training and selection

We train the same three classifiers as done in Task 1: Random Forest, Logistic Regression, and for each classifier we run a grid search with 5-fold cross-validation, where the folds are also split by subject (using GroupKFold), so the model is always validated on subjects it has not seen during training.

Classifier	Hyperparameter	Values searched
Logistic Regression	Scaler	none, Standard, Robust, MinMax, **PowerTransformer**
	C	**0.01**, 0.1, 1, 10
	20 configurations	
SVM	Scaler	Standard, Robust, **PowerTransformer**
	Kernel	linear, **RBF**, polynomial
	C	0.01, 0.1, 1, **10**
	γ (RBF only)	0.01, 0.001, **0.0001**, auto
	Degree (polynomial only)	2, 3, 4
	96 configurations	
Random Forest	Number of trees	100, **200**
	Maximum depth	3, 5, 8, **10**, 15, 20, unlimited
	Minimum samples per leaf	1, **2**, 5, 10, 20, 50
	84 configurations	

Table 4.1.1: Hyperparameter tuning grid for KDEF: Selected values (highest macro-F1) are in bold

4.2 Results

	FER2013		KDEF	
Classifier	Validation	Test	Validation	Test
Logistic Regression	0.395	0.378	0.669	0.581
SVM	**0.505**	**0.568**	**0.679**	0.604
Random Forest	0.440	–	0.611	**0.614**

Table 4.2.1: Macro-F1 of models trained on HOG features of KDEF

Table 4.2.1 shows that all three classifiers score higher on KDEF than on FER2013, even though they are trained on only 784 images instead of about 28,700. The reason is that KDEF is a much cleaner dataset. Every photo is taken in the same studio, with the same lighting and camera position, and the actors show each emotion clearly, so the faces line up well and the expression is almost the only thing that differs between images. The HOG features of two happy faces therefore look very similar, and the classifiers need few examples to learn each emotion. FER2013 images come from the web, with different lighting, head poses, faces partly covered by hands or glasses, and labels that are sometimes wrong, so the classifiers have to learn the same emotion from very different-looking images.

4.3 Assignment Questions

Why are KDEF images easier or harder to classify? Overall, KDEF is easier: with the same HOG features, the SVM reaches a macro-F1 of 0.679 in cross-validation and 0.604 on the test subjects. As mentioned before, the images in the KDEF dataset are consistent and is balanced. Each facial attribute is present in a similar pixel vicinity across all images of an emotion. This makes the models learn better, unlike the FER2013 where there is no consistency in the subjects present, alignment and facial features/attributes of the face etc.

To understand how these alignment of these facial features could be of concern for the model, we generated the mean face for each emotion in both datasets. This was done to understand how well positioned each facial feature is in all images across the dataset. The more blurry the images are, the more inconsistent the images are in the dataset for each emotion.

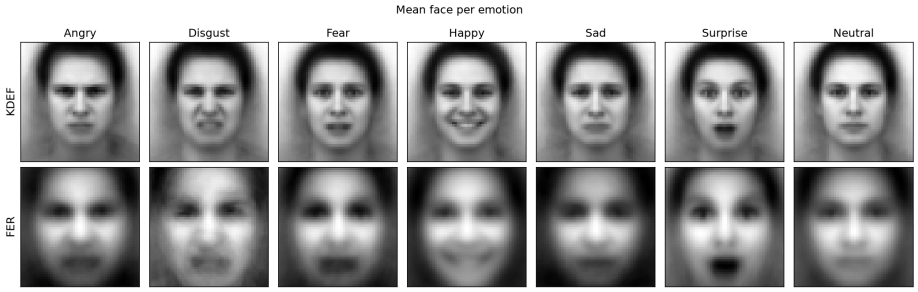


Figure 4.3.1: Mean face per emotion of KDEF and FER2013 training images

Which models change ranking?

Rank	FER2013 validation	KDEF cross-validation	KDEF test
1	SVM (0.505)	SVM (0.679)	Random Forest (0.614)
2	Random Forest (0.440)	Logistic Regression (0.669)	SVM (0.604)
3	Logistic Regression (0.395)	Random Forest (0.611)	Logistic Regression (0.581)

Table 4.3.1: FER2013 vs KDEF model comparison

On KDEF, Logistic Regression jumps from last to second place in validation, while the Random Forest falls to last. We think this is because the well-aligned KDEF faces make the HOG features almost linearly separable, so a strongly regularized linear model ($C = 0.01$) works well with few training images. Although, the Random Forest performs better in the test set by a small margin, we choose the best model from the validation set.

Which emotions transfer well between datasets? To see which emotions hold up on both datasets, we compare the F1 score per emotion of the KDEF models with that of the FER-trained SVM from Task 1 (Table 4.3). Happy is the easiest emotion on both datasets (0.93 on KDEF and 0.75 on FER2013 for the SVM), and Surprise also scores well on both (0.62 and 0.73). Both expressions have large, distinctive shapes, a smile or an open mouth with raised eyebrows, which HOG picks up whether the expression is posed in a studio or caught in a web photo. Disgust scores well on both too, but FER2013 contains very few Disgust images (1.5%), so this result is less certain. Fear, Sad and Angry are among the hardest emotions on both datasets, with F1 scores between 0.33 and 0.55.

	KDEF			FER2013
Emotion	Logistic Regression	SVM	Random Forest	SVM
Angry	0.50	0.50	0.65	0.44
Disgust	0.53	0.65	0.69	0.62
Fear	0.41	0.33	0.34	0.46
Happy	**0.88**	**0.93**	**0.77**	**0.75**
Sad	0.51	0.55	0.53	0.43
Surprise	0.63	0.62	0.63	0.73
Neutral	0.61	0.66	0.69	0.55

Table 4.3.2: F1 score per emotion - classifier comparison

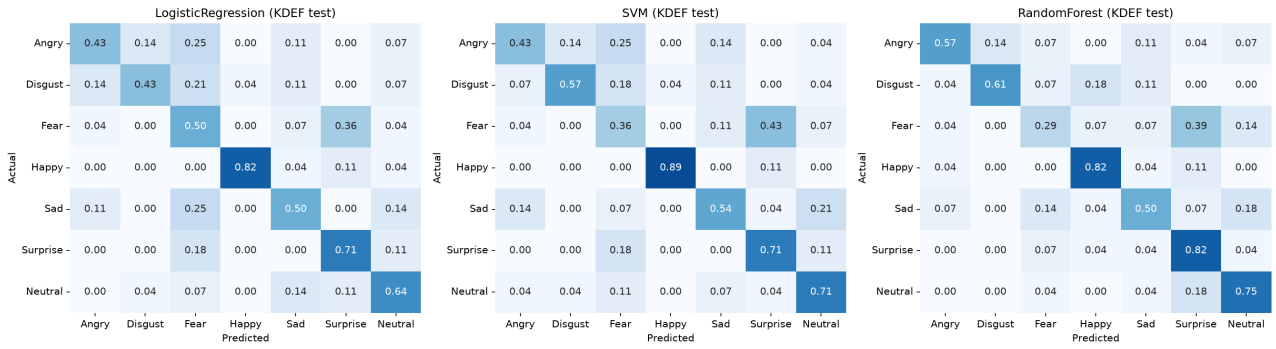


Figure 4.3.2: Confusion matrices(normalized values) for KDEF dataset

The confusion matrices in Figure 4.3.2 show that most mistakes happen between emotions that look alike. Fear is the clearest example: the SVM mistakes more fearful faces for Surprise (43%) than it gets right (36%), and the other two classifiers do the same. This is not surprising, since both expressions have wide-open eyes, raised eyebrows and an open mouth. In the same way, Sad is often mistaken for Neutral (21%), and Angry for Fear (25%) or Disgust (14%). Based on this, we expect Happy and Surprise to transfer best between the datasets, and Fear, Sad and Angry to transfer worst.

5 Task 3: Cross-dataset generalization (20%)

Test both directions required by the assignment protocol: **FER→KDEF** (FER-trained model applied to cropped/preprocessed frontal KDEF) and **KDEF→FER** (retrain the KDEF-selected configuration on its training subjects, then evaluate on FER). Keep matching 48×48 grayscale normalization and class mapping throughout; do not adapt the test-domain examples using ground-truth labels. Keep FER PrivateTest untouched until final evaluation.

Table 2: Within- and cross-dataset performance, on explicitly stated held-out sets.

Train → evaluation	Accuracy	Bal. acc.	Macro-F1
FER → FER (PrivateTest)	–	–	–
KDEF → KDEF (test subjects)	–	–	–
FER → KDEF (test subjects)	–	–	–
KDEF → FER (PrivateTest)	–	–	–

[Replace: List any model/input adaptations, define domain shift, quantify transfer degradation against in-domain comparisons, analyze per-emotion confusion, and explain which features transfer best and why. Note crop/alignment, resolution and grayscale/RGB as confounders.]

6 Task 4: Explainable fuzzy classifier (10%)

Features and protocol. We extracted seven geometric features from the 68 dlib facial landmarks: mouth width, mouth opening, left and right eye opening, eyebrow height, mouth width-to-height ratio (*smile ratio*), and mouth-corner asymmetry. Most distances were normalized by face width, while eye opening was normalized by eye width. All 980 frontal KDEF images were processed successfully, without landmark detection failures.

The predefined subject-independent split contained 56 training subjects (784 images) and 14 held-out test subjects (196 images). Within the training partition, 44 subjects (616 images) were used for fitting and 12 (168 images) for validation. MinMax scaling was fitted only on the corresponding training partition.

Fuzzy model. We trained seven one-versus-rest Takagi–Sugeno fuzzy models with PyFUME, one for each emotion. Each model uses Gaussian membership functions in its rule antecedents and a learned linear consequent. The numerical outputs are combined through fuzzy inference, and the emotion with the highest output score is selected. These scores are not calibrated probabilities.

We compared two and three clusters per emotion using macro-F1 on validation subjects. The three-cluster configuration was selected before testing and subsequently refitted on all 784 training images.

Table 3: Subject-independent KDEF fuzzy classification results.

Split	Clusters	Images	Acc. (%)	Bal. acc. (%)	Macro-F1
Validation	2	168	71.43	71.43	0.6881
Validation	3	168	73.21	73.21	0.7087
Final test	3	196	63.27	63.27	0.6082

Rule interpretation. The final system contains 21 fuzzy rules, three per emotion. A representative rule from the happiness model is:

> *Happy, Rule 1 (abridged):* IF mouth width belongs to cluster 1 (Gaussian centre 0.762) AND mouth opening belongs to cluster 1 (centre 0.204), together with the remaining geometric conditions, THEN calculate the happiness score using the fitted linear consequent.

This rule dominated 94.6% of happy training images but only 2.2% of other images. Its relatively wide-mouth configuration is consistent with smiling. The strongest rules for fear and sadness showed weaker separation, and overlapping membership functions limit their linguistic interpretation.

Evaluation and limitations. On unseen subjects, happiness (28/28), surprise (26/28) and disgust (23/28) achieved the highest recall. Fear (7/28) and sadness (8/28) were more difficult: 14 fearful faces were classified as surprise, and 12 sad faces as neutral. The roughly ten-percentage-point accuracy decrease from validation to testing indicates limited generalization across individuals. Although the fuzzy antecedents are inspectable, the linear consequents and interactions between the seven models complicate explanations of individual predictions.

7 Task 5: Creative challenge (25%)

Choose and justify **at least two** FER2013 improvements: hyperparameter tuning, feature engineering/selection, class balancing, augmentation, data curation, label verification, KDEF-derived geometric information, emotion prototypes, or domain-invariant representations. *[Replace: Name the actual two or more changes, implementation details, expected mechanism, and risks.]*

Table 4: FER2013 controlled ablation: select with PublicTest, evaluate the final choice once on PrivateTest.

Variant (PublicTest)	Accuracy	Bal. acc.	Macro-F1
Original baseline	–	–	–
Improvement A only	–	–	–
Improvement B only	–	–	–
Combined A + B	–	–	–

[Replace: Quantify gains/losses and tradeoffs. Explain class-wise and failure-case changes, computational cost, potential leakage, and one final untouched PrivateTest comparison. Reference appendix matrices.]

8 Task 6: Real-time webcam demo (10%)

Adapt the best selected model into a command-line executable Python file that opens the laptop webcam and performs live emotion recognition. Preserve the exact training face crop, alignment, feature extraction, normalization and class mapping in the inference path.

[Replace: State demo command, chosen model/.pkl, hardware/OS, measured or observed FPS/latency (if available), stability, real-world behavior, lighting/pose/occlusion issues and what was fine-tuned. Explain whether the model performed well enough, and why.]

9 Discussion, limitations and conclusion

[Replace: Synthesize: best features/classifiers on each dataset; relative difficulty of each emotion; changes of ranking; strongest and weakest transfers; fuzzy interpretability versus performance; benefits and costs of creative improvements; webcam readiness. Contrast dataset-specific and common limitations. Draw conclusions only from actual measurements and concrete error examples.]

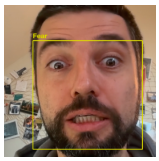
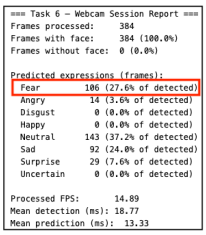
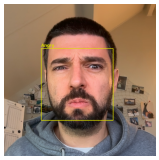
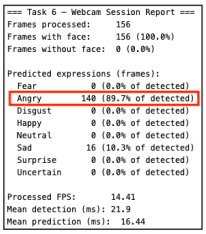
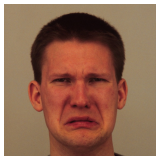
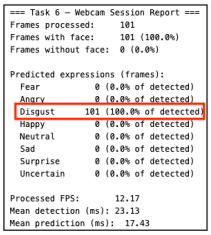
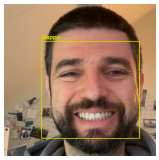
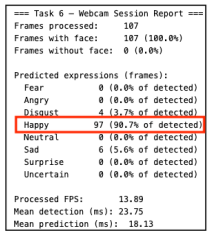
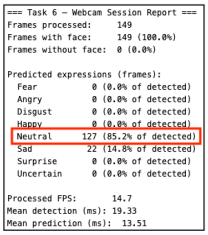
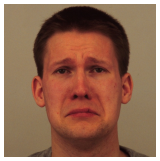
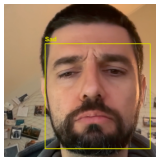
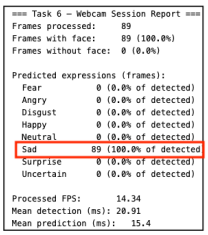
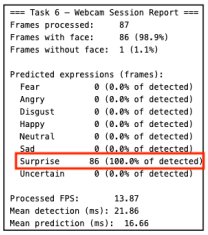
Posed expression	KDEF reference	Webcam capture	Prediction
Fear			
Angry			
Disgust			
Happy			
Neutral			
Sad			
Surprise			

Table 5: Qualitative comparison of posed KDEF expressions, corresponding webcam expressions, and fuzzy classifier predictions. KDEF subject M11, acquisition A, frontal pose. This illustrative experiment does not establish classification accuracy.

Declaration of AI tool use

During the preparation of this work, I/We used [NAME TOOL / SERVICE / VERSION OF AI TOOL] in order to [REASON]. The following parts of the assignment were affected/generated by AI tool usage: [INTRODUCTION / METHODS / xxx, DISCUSSION]. After using this tool/service, [NAME STUDENT(S)] evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, [NAME STUDENT(S)] take(s) full responsibility for the content of their work.

References

Goeleven, E., De Raedt, R., Leyman, L., & Verschuere, B. (2008). The Karolinska Directed Emotional Faces: A validation study. *Cognition & Emotion*, *22*(6), 1094–1118. https://doi.org/10.1080/02699930701626582

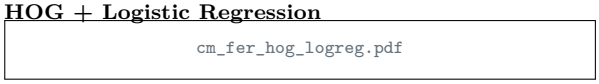
Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., … Bengio, Y. (2013). Challenges in Representation Learning: A Report on Three Machine Learning Contests. In M. Lee, A. Hirose, Z.-G. Hou, R. M. Kil, D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi, & G. Weikum (Eds.), *Neural Information Processing* (pp. 117–124, Vol. 8228). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-42051-1_16

A Confusion matrices for reported models

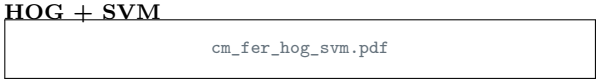
[Replace: Replace every relevant slot with an actual confusion matrix exported by the corresponding notebook. Remove unreported model configurations; include matrices for all reported final models, with evaluation split, class ordering, count or normalization labeled.]

A.1 FER2013 baseline models (PublicTest; final chosen model on PrivateTest)

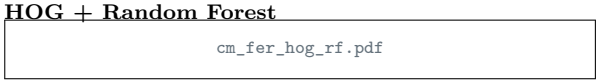
HOG + Logistic Regression



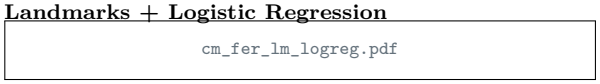
HOG + SVM



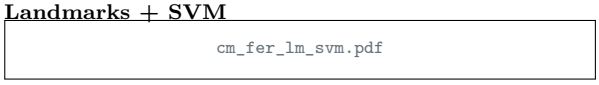
HOG + Random Forest



Landmarks + Logistic Regression



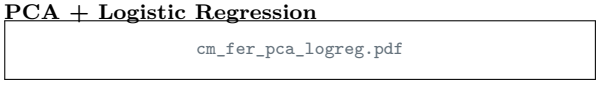
Landmarks + SVM



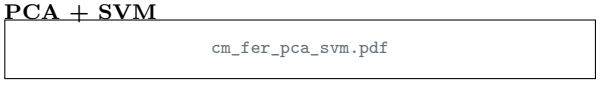
Landmarks + Random Forest



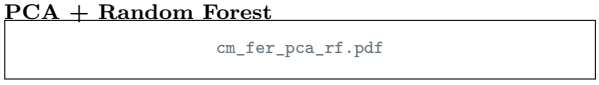
PCA + Logistic Regression



PCA + SVM



PCA + Random Forest



FER final selected (PrivateTest)



A.2 KDEF, cross-dataset, fuzzy and creative models

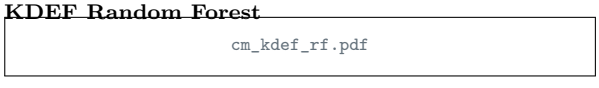
KDEF Logistic Regression



KDEF SVM



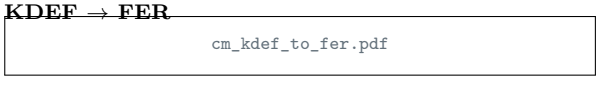
KDEF Random Forest



FER → KDEF



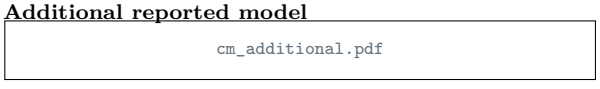
KDEF → FER



Creative variant(s)



Additional reported model



A.3 Fuzzy classifier: held-out KDEF subjects

Table 6: Final fuzzy classifier confusion matrix (counts; 196 test images from 14 unseen subjects). Rows are true labels and columns are predictions.

True / Pred.	Af	An	Di	Ha	Ne	Sa	Su
Afraid	7	1	2	0	2	2	14
Angry	0	14	10	0	2	2	0
Disgust	0	3	23	1	0	1	0
Happy	0	0	0	28	0	0	0
Neutral	0	3	0	0	18	6	1
Sad	0	6	2	0	12	8	0
Surprise	0	0	0	0	1	1	26

Af: afraid; An: angry; Di: disgust; Ha: happy; Ne: neutral; Sa: sad; Su: surprise.